Project Plan:

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Name	Type	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10
Reva Agrawal	Goals		Data cleaning / pre-processing	Algorithm dev.			PE evaluation (+ more dev. if needed)	Work on website/report -- results			
	Deliverables		Access the intel telemetry data using globus and sample it	Update the report regularly with progress on methods, modelling and discussion			Evaluate the model performances and include results to the report.	Completed results section			
Jordan Lambino	Goals		Data cleaning / pre-processing	- Create website, add basic info	- Create report with basic info - Sync with group on dev. process - Design website format/theme	- Generate plots for model analysis - Sync with group on dev. process	- Add data cleaning, algorithm dev. details to website	Work on website/report -- discussion & conclusion			
	Deliverables		Explanation of the queries and the output	- Link to website with introduction	- Create report with basic info - Sync with group on dev. process - Draft of website format/theme	- Completed charts, added to website	- Link to website with added sections	Completed conclusion section			
Dhruv Patel	Goals		Data cleaning / pre-processing	Algorithm dev.			PE evaluation (+ more dev. if needed)	Work on website/report -- methods			
	Deliverables		Summary Statics of Sample data, and running the queires	Run the ouput mechanism for any errors and at the end of week 5 should be able to produce complete synthetic data			Evaluate the model performances	Completed methods section			

Week Number: 8

Outside of your meeting with your mentor, when did your group meet this week?

Friday, February 27, 2026

Time: 2-3 pm

Location: Zoom

Attendees: All group members (Reva Agrawal , Jordan Lambino, Dhruv Patel)

What did each group member do this week? Where did they get stuck?

- **Dhruv: This week, I focused on refining our model's performance by systematically optimizing the decision thresholds used in our evaluation pipeline. After running a series of**

controlled experiments on the subsample, I identified threshold settings that significantly improved both precision compared to our earlier baseline. Most of my time went into analyzing how different thresholds interacted with the distribution of our model outputs, validating improvements across multiple subsets of the data, and ensuring that the gains were consistent rather than overfit to a single slice. I also documented the full threshold-tuning process so we can integrate it cleanly into the larger pipeline once the team finalizes the next stage of our workflow. This work positions us well for scaling the evaluation to the full dataset once the remaining components are in place.

- **Reva: This week I worked on reproducing our results for different query releases. I set up the workflow so that we first run to regenerate the baseline CSV files and then run either gaussian mechanism or laplace mechanism with the database. The output summaries are automatically written into the corresponding data folders based on the mechanism, database, and epsilon value. This helps us quickly see what works best without digging through the code. Since we are each responsible for two query releases, I also started building the mathematical expressions behind my queries before releasing the differentially private results. I focused on clearly defining the non private function and thinking through the sensitivity so I can apply the correct DP mechanism and then evaluate the output properly.**
- **Jordan: This week I made updates to the website including slight changes to format/layout and edits to the body of the website. I encountered a particular roadblock when trying to add an auto-scroll feature for each section of the website. I plan on working with my group to address this issue or come up with a better alternative. Aside from website design, I reviewed Dhruv's documentation of the DP query release system, particularly in preparation for the query release we plan to perform this week. Lastly, I added information regarding DP query release to our website.**

What will each group member do next week?

- **Dhruv:Next week, I will continue refining the threshold-optimization work by validating the improved settings across additional subsets of the data and ensuring that the gains we observed hold under different sampling conditions. I also plan to integrate the optimized thresholds directly into our evaluation pipeline so they become part of the end-to-end workflow rather than a separate experimental step. As part of this integration, I'll add automated checks that compare model performance under multiple threshold candidates, allowing us to track stability as we scale to the full dataset. Once this is in place, I'll begin coordinating with the rest of the team to align the threshold logic with the broader model-development and reporting components, ensuring that our improvements translate cleanly into the final system.**
- **Reva: Next week I will finish cleaning and finalizing the pipeline and release two queries in a fully structured, differentially private way. I will also work on writing the report sections for each query with the formal mathematical explanation and privacy reasoning. Along with that, I will help finalize the poster and update the website so everything clearly reflects our query release approach and evaluation setup.**

- **Jordan: Next week I will add additional features to the website for accessibility, including hyperlinks for each section, images/visuals, and potentially dashboards to display statistics. I also plan to update the website with updated information based on our progress from the past week. By the end of the week, we should have the majority of results finalized, which I will then add to the website. Along with Reva, I plan to release 2-4 queries following our DP pipeline, and report the results for analysis. In addition to query release, I plan on adding results/analysis to the report and poster.**

As a group, do you still feel on track to execute your project proposal?

- **Dhruv: Yes.**
- **Reva: Yes! I am confident we will execute it well and on time!**
- **Jordan: Yes, we need to have steady progress but I am confident.**

Week Number: 5

Outside of your meeting with your mentor, when did your group meet this week?

Friday, February 6, 2026

Time: 2-3 pm

Location: Zoom

Attendees: All group members (Reva Agrawal , Jordan Lambino , Dhruv Patel)

What did each group member do this week? Where did they get stuck?

- **Dhruv:We began constructing our data pipeline using the subsample so we could iterate on our workflow more efficiently. A significant portion of my time went into developing the private-evolution code required to support this pipeline and ensure it could scale to the full dataset. As we progressed, the subsample also exposed several missing tables and pockets of incomplete data that we hadn't anticipated, prompting us to think more systematically about how to handle these gaps. To address missing tables, we started outlining a set of fallback and validation strategies. This included building checks that verify table availability before each pipeline stage runs, defining clear rules for when to halt execution versus when to proceed with partial data, and designing placeholder structures that allow downstream processes to continue without breaking. For incomplete or sparsely populated tables, we explored approaches such as schema-level consistency checks, automated reporting to flag unexpected null patterns, and logic for imputing or reconstructing**

essential fields when possible. These early investigations helped us understand the reliability of the full dataset and informed the robustness requirements for the production-scale pipeline.

- **Reva: This week, following the professor's guidance, we shifted our focus to working with a smaller subsample of the data, around 100–300 MB, since loading the full dataset was taking too long. We started building a data pipeline using this subsample so that we could develop and test our workflow more efficiently. Much of my time was spent writing the private evolution code to support this pipeline. While working on this, we also started thinking about strategies for dealing with missing tables and incomplete data, as the subsample revealed some gaps we hadn't anticipated in the full dataset.**
- **Jordan: This week I continued working on the website, adding to the Introduction and Methods sections. I also brainstormed layout ideas which best communicate our project with my group. I've still encountered the same roadblock regarding data visualization given the confidential nature of the dataset. In addition to the website and report, I helped out with EDA roadblocks and analyzed preliminary results.**

What will each group member do next week?

- **Dhruv:Next week, my priority will be to take the plan we finalize as a team and translate it into a concrete, executable workflow. This includes refining and optimizing the current pipeline so it runs smoothly on the subsample, and then implementing the agreed-upon strategies for handling missing tables and incomplete data. Once the approach is locked in, I will begin integrating each component directly into the pipeline—whether that involves imputation logic, schema-validation checks, fallback mechanisms, or automated reporting. After the subsample pipeline is stable and fully aligned with the finalized plan, we will begin scaling the system to support the full 3.5 TB dataset. During this phase, I will continue strengthening the private-evolution code, incorporating any adjustments identified during testing, and ensuring that every step is documented clearly. The goal is to produce a fully reproducible, end-to-end workflow that can transition seamlessly from the subsample to the full dataset without unexpected failures.**
- **Reva: Next week, I will focus on refining and optimizing the pipeline so it runs smoothly on the subsample, while also experimenting with different imputation methods for the missing tables we identified. Once the pipeline works reliably on the smaller dataset, we plan to scale it up to handle the full 3.5 TB dataset. I will continue improving the private evolution code, integrating any necessary adjustments, and documenting the process so that we have a reproducible workflow when we transition to the full data.**
- **Jordan: Next week, I plan to add details to the website about data collection and cleaning, showing our process and methods. Additionally, I will add some information regarding algorithm development. I plan to have a more refined website layout/design, along with some progress on the data visualization aspects. With Reva's help, I will add relevant information to the report. In parallel, I plan to help out with model development and evaluation when needed.**

As a group, do you still feel on track to execute your project proposal?

- **Dhruv: Yes! We have a decent amount of the pipeline planned. As long as we get the other 5 percent done in the first three days, we will be good to go!**
- **Reva: Yes! If we can have the pipeline ready for the subsample, we will be good to go!**
- **Jordan: Yes, we need to make decent progress throughout this next week, but we're on track for success so far.**

Week Number: 4

Outside of your meeting with your mentor, when did your group meet this week?

Friday, January 30, 2026

Time: 2-3 pm

Location: Zoom

Attendees: All group members (Reva Agrawal , Jordan Lambino, Dhruv Patel)

What did each group member do this week? Where did they get stuck?

- **Dhruv: This week, I worked on setting up the database infrastructure for our project. I organized all 16 source tables (in parquet and .gz formats) into a consolidated database_tables folder. I then created a Python script to build a DuckDB database that loads all tables into two schemas (university_analysis_pad and university_prod) to match our SQL reporting structure. After that, I ran the SQL script that generates the reporting table addressing the project's main bottleneck. To compute both the upper bound (full utility) and the lower bound (full privacy), we need to run SQL queries that produce the corresponding outputs. This week, we successfully obtained the result for the upper bound, which represents the full-utility scenario.**
- **Reva: This week, I focused on merging all of our different tables to form the large dataset we plan to use for EDA. The dataset is massive now, with over 5 billion rows, so managing the joins and ensuring consistency across tables was a challenge. I also coordinated with the other B15 group to divide the SQL queries, and we took responsibility for the first 12 queries. Much of my time was spent checking that the merged tables preserved all relevant columns and metrics, and that no data was lost or misaligned during the merging process. I ran into a few issues with overlapping columns and inconsistencies in table formats, which required carefully reconciling them, but overall the dataset is now ready for analysis.**
- **Jordan: I continued working on the website this week and drafted different organizational ideas. I also worked on the report and cleaned up the introduction section in alignment with**

the revised project direction (compared to the project proposal). Along with my group, I discussed with the other B15 group and split the queries for future analysis. Regarding the website, I've encountered a general roadblock concerning interactivity. I looked at some previous capstone websites (particularly from last year's B15 equivalent), and wasn't quite sure how they incorporated certain features.

What will each group member do next week?

- **Dhruv: Next week, I will run the SQL queries on the complete dataset with full privacy settings (maximum privacy/no privacy budget limits) to establish the lower bound limit for our differential privacy implementation. This will help us understand the distribution of noise and serve as a baseline for comparison when we apply differential privacy with varying epsilon values. This lower bound will be critical for evaluating how much noise we need to add while maintaining data utility.**
- **Reva: Next week, I will begin running EDA on the merged dataset to explore distributions, identify outliers, and understand the relationships between key variables. I will also execute the baseline code to establish reference performance metrics. In parallel, I plan to draft a rough version of the report, summarizing the dataset construction process, initial observations from the EDA, and the steps taken so far. As the analysis progresses, I will update the report with new findings and visualizations, gradually building a comprehensive record of our work and results.**
- **Jordan: I plan to run EDA along with my group members and assist with roadblocks when needed. I will continue working on both the website and report in preparation for the upcoming checkpoint. With regard to visualizations for the website and report, I will brainstorm with my team and come up with different ways we can show our results, given that the telemetry data is confidential. For the website overall, I need to discuss with my team regarding theme, layout, etc. As we progress with our analysis, I will add relevant information and findings.**

As a group, do you still feel on track to execute your project proposal?

- **Dhruv: Yes! The EDA is almost over with running of upper bound limit next week lower bound and the model.**
- **Reva: Yes! We have substantial progress on the dataset creation this week which is great!**
- **Jordan: Yes! Hopefully, the EDA goes smoothly this week, which would position us well for algorithm development in the coming weeks.**

Week Number: 3

Outside of your meeting with your mentor, when did your group meet this week?

Friday, January 23, 2026

Time: 2-3 pm

Location: Zoom

Attendees: All group members (Reva Agrawal , Jordan Lambino, Dhruv Patel)

What did each group member do this week? Where did they get stuck?

- **Dhruv: This week, I began by downloading and setting up the dataset locally to ensure I could reliably access and load the raw data without issues. I inspected the structure of the raw files, including columns, data types, and metadata, to understand how the data was organized and what preprocessing would be required. Before starting exploratory data analysis (EDA), I focused on the ELT process, working on extracting the data, loading it into a usable format, and planning necessary transformations to clean and standardize the dataset. This included checking for missing values, inconsistent formats, and understanding how users, events, and timestamps are represented. I also began outlining an EDA plan, but realized that meaningful analysis depends on having a clean and stable transformed dataset, so prioritizing ELT was necessary. I did not encounter major technical blockers, but ensuring the data was properly structured before analysis took more time than expected.**
- **Reva: This week, I worked on reviewing all of our SQL queries and mapping each one to the exact tables and columns needed to run them correctly. My main task was making sure no metrics, join keys, or filter conditions were missed, especially for weighted averages and time-based joins. I also checked our schema design assumptions to confirm that the proposed tables support the analyses we want to run. I got stuck mainly because of the large number and complexity of the queries, since many of them used multiple tables with small but important dependencies, which meant I had to review them several times. Progress was slower than expected because accuracy mattered more than speed, but this helped catch potential gaps early.**
- **Jordan: This week, I created the website and report to track our group's progress. I added introductory details, mainly revolving around context from a high-level perspective. There were few roadblocks for this task since I had MAs from the previous quarter to use as references. I also reviewed Reva's schema exploration for accuracy to avoid future roadblocks. I briefly discussed how queries were to be divided with the other B15 group, but we are still analyzing the queries before making a final decision.**

What will each group member do next week?

- **Dhruv: Next week, I will continue refining the ELT pipeline to ensure the transformed dataset is consistent, well-documented, and ready for analysis. Once the ELT process is**

complete, I will proceed with exploratory data analysis (EDA), focusing on distributions, summary statistics, and identifying potential outliers or patterns that may affect downstream analysis and privacy considerations. I will use insights from EDA to help inform which analytical queries are feasible within the project's differential privacy constraints and to support the group's decisions around query selection and scope.

- **Reva: Next week, I will start by running the baseline analyses using DP-VAE once the specialized schema is loaded on S3. In parallel, I will begin writing the full report, including the introduction, background, problem setup, and methodology. As results come in, I will connect the analysis back to the motivation and assumptions behind the schema, and gradually build out the rest of the report around those findings in the coming weeks.**
- **Jordan: Next week I will continue adding to the website and report to keep up with our group's progress. Regarding the website, I plan to update the format and design as needed to effectively communicate our work. I also plan to sync with the other B15 group and finalize the division of queries. Following the project plan above, I need to draft high-level explanations of the relevant queries along with the desired outputs. In parallel, I will assist other group members with roadblocks when needed.**

As a group, do you still feel on track to execute your project proposal?

- **Dhruv: Yes! But there is a need to complete ELT and EDA fast so that the generation pipeline can be completed on time.**
- **Reva: Yes! We figured out one of our major bottlenecks this week so progress looks steady!**
- **Jordan: Yes! We have to catch up slightly due to the slow start but 100% confident that our plan can be executed.**

Week Number: 2

Outside of your meeting with your mentor, when did your group meet this week?

Friday, January 16, 2026

Time: 2-3 pm

Location: Zoom

Attendees: All group members (Reva Agrawal , Jordan Lambino, Dhruv Patel)

What did each group member do this week? Where did they get stuck?

- **Dhruv: I initially attempted to download the dataset using the local machinery and Globus Connect, but the transfer process was taking significantly longer than expected due to the dataset's size and transfer overhead. To avoid blocking progress, I shifted focus to setting up DuckDB as an alternative, allowing me to work directly with the data more efficiently and begin querying and exploratory analysis without waiting for the full download to complete.**
- **Reva: I spent this week working on accessing the Intel telemetry data and figuring out how to work with it at scale. Instead of trying to load everything at once I began breaking the data into smaller pieces and identifying which fields are actually needed for the analyses we care about. I started outlining a compact schema that keeps only the relevant columns needed to support the SQL queries. Progress was slower than expected because of the dataset size.**
- **Jordan: This week I worked on collecting the telemetry data from Globus and narrowing our project scope. With regard to data collection, I encountered a roadblock due to the dataset size. We decided to batch-process the data as a temporary solution, though our methods are still being refined. To address the issue of project scope, I coordinated with our mentor who decided that we could split up the queries however we'd like among groups. The final query assignments are still WIP.**

What will each group member do next week?

- **Dhruv: Set up and validate a DuckDB-based data pipeline to efficiently ingest and query the dataset, ensuring correct schema alignment and reproducibility. Use DuckDB to perform exploratory data analysis (EDA), including summary statistics, distributions, missingness patterns, and key correlations that inform modeling decisions. Based on insights from the EDA, implement and evaluate a baseline non-private model (with clearly defined metrics) to establish a performance reference point against which differentially private and synthetic data models can be compared.**
- **Reva: Next week I will continue processing the data in batches and finalize a reduced schema that supports the required SQL queries. Our professor also shared access to a repository from last year's project, so I plan to study that code to understand how previous teams loaded and structured the data and then build on those ideas to improve our pipeline. Once that is done, I'll move forward with the EDA.**
- **Jordan: I plan to continue working with my group members to batch-process the telemetry data, assisting with roadblocks when needed. Once our data collection pipeline is more refined, I will execute EDA to gain some preliminary insights about the data we're working with. I will also work with the other B15 group to divide the queries accordingly, and then draft explanations of the assigned ones. In alignment with the Project Plan above, I will create our group's website and include introductory details.**

As a group, do you still feel on track to execute your project proposal?

- **Dhruv: Yes**

- **Reva: Yes I do. While working with the full dataset has taken some adjustment, we now have an approach discussed with our mentor that keeps us aligned with the overall goals.**
- **Jordan: Yes. We worked with our mentor to develop a reasonable plan given both the size of the dataset and the number of queries being dealt with. Though these first two weeks have been slightly more complicated than expected, we now have a narrowed scope to focus on.**

Week Number: 1

Outside of your meeting with your mentor, when did your group meet this week?

Friday, January 9

Time: 2-3 pm

Location: Zoom

Attendees: All group members (Reva Agrawal , Jordan Lambino, Dhruv Patel)

What did each group member do this week? Where did they get stuck?

- **Dhruv:** Focused on dataset access and infrastructure setup. After encountering slow transfer times using local machinery and Globus Connect, Dhruv set up DuckDB to enable efficient querying and inspection of the large dataset. He verified that the data could be loaded and explored without issues and did not encounter major technical blockers; the main challenge was assessing how to handle the dataset's scale within the project timeline.

- **Reva:** Concentrated on understanding the dataset structure and contents. Reva reviewed the available fields, data types, and documentation to assess feasibility for analysis and identify potential sensitivity concerns relevant to differential privacy. She did not face technical obstacles but noted that the breadth of possible analytical queries could make the project overly ambitious without scope reduction.

- **Jordan:** Led discussions on project scoping and alignment. Jordan evaluated the overall size and complexity of the dataset and helped identify that the full set of potential queries would likely be too broad for a single quarter. His main challenge was not implementation-related, but rather determining how to prioritize or narrow queries while preserving the core goals of the project.

What will each group member do next week?

- **Dhruv:** Will begin exploratory data analysis (EDA) using DuckDB, focusing on basic summaries, distributions, sparsity patterns, and computational feasibility. He will also prepare initial statistics that can later be used to evaluate utility and privacy trade-offs.

- **Reva:** Will develop preliminary visualizations and descriptive summaries to better understand data distributions and identify candidate queries suitable for differentially private analysis. She will also start thinking about which metrics best capture utility preservation.

- **Jordan:** Will coordinate with the professor to clarify expectations around query scope and confirm which subset of queries should be prioritized. He will also help translate these decisions into a concrete analysis plan for the group.

As a group, do you still feel on track to execute your project proposal?

- **Dhruv:** Yes. Dataset access and tooling are in place, and early identification of scope issues allows for informed adjustments before heavy implementation begins.

- **Reva:** Yes. The group has aligned on goals and is moving into EDA with a clear understanding of constraints, which supports a realistic and focused execution plan.

- **Jordan:** Yes. By addressing scope early and planning to confirm expectations with the professor, the project remains well-positioned to meet its objectives within the quarter.